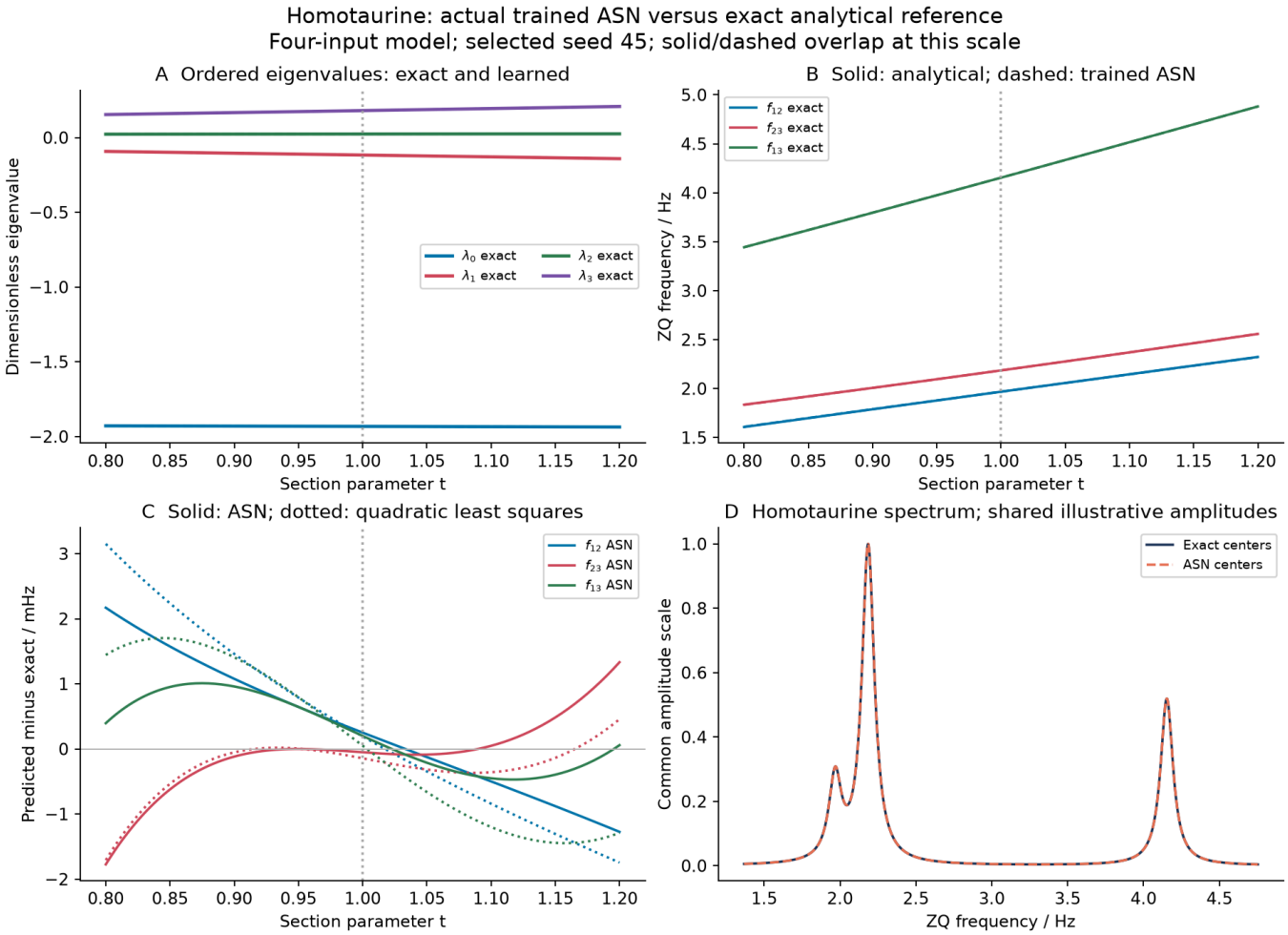
Homotaurine: analytical solution and formulas learned by ASN

Real-molecule application for the revised manuscript · 18 October 2026

Both results are provided. The exact analytical construction is the reference; the other formulas are extracted from an actually trained `nmrkan.models.KharKAN`, the activation-selection network (ASN) used in the manuscript. Supervised targets come from numerical 4×4 diagonalization, not from the analytical cubic. This repeats the DSS protocol with the supplied homotaurine reference and a new local training distribution.

Line	Exact q=0 / Hz	Trained ASN / Hz	Error / mHz	Exact fitted q, even / Hz
f12	1.967922603	1.968173663	+0.251060	1.951897472
f23	2.185439927	2.185388866	-0.051062	2.203534522
f13	4.153362531	4.153562529	+0.199998	4.155431994

The selected local model has held-out frequency RMSE **0.002135162 Hz**, maximum line error **0.02108312 Hz**, and **0** ordering violations on 10,000 test points. These are surrogate errors against the nearest-neighbor secular model. They are not residuals against measured peaks. The last table column shows a separate physical correction, which the four-input q=0 ASN was not trained to predict.



1. Molecular parameters and the experimental connection

The supplied `ASN_DSS_homotaurine_Hamiltonian.pdf` and `asn_three_ch2.py` specify

$(J_A, J_M, J_X, d, e) = (-12.743923, -13.969277, -14.109177, 3.093406, 2.344390)$ Hz,

$q = J_{AX} - J_{AX'} = 0.027306$ Hz.

Here d and e are full coupling differences; off-diagonal matrix elements are d/2 and e/2. The source attributes this row to a 15 August 2026 email. It explicitly flags a filename labeled 298 K but a body labeled 289 K, and an unreconciled subsequent revision. We retain the internally consistent physical row as an illustrative molecular reference. Its temperature and final-fit status remain unresolved. The raw spectrum and source emails are not available here.

The experimental precedent is Sheberstov, Sonnefeld and Bodenhausen, J. Chem. Phys. 160, 144308 (2024). The authors' accepted manuscript, Fig. 9b,d and Experimental section describes homotaurine ZQ sections with in-phase DQ-AM-SLIC preparation and SQ-A-SLIC or SQ-X-SLIC reconversion. It reports measurements at 298 K and 500 MHz on a D_2O mixture containing approximately 50 mM of each molecule. This establishes the relevant experiment class; it does not resolve the temperature of the supplied 2026 parameter row. The published ZQ linewidth scale is about 0.1 Hz, which we use for illustrative Lorentzians. No experimental intensities or fit uncertainties are inferred from it.

2. Hamiltonian and what can be neglected

For $S_J = J_A + J_M + J_X$, the even block in the basis (TSS, STS, SST, TTT) is

$$H_+ = -\frac{3S_J}{4} I + \begin{pmatrix} J_A & d/2 & q/2 & e/2 \\ d/2 & J_M & e/2 & q/2 \\ q/2 & e/2 & J_X & d/2 \\ e/2 & q/2 & d/2 & S_J \end{pmatrix}.$$

Hamiltonians are in Hz and time evolution is $\exp(-i2\pi H\tau)$. The common pair chemical-shift and mean interpair secular terms vanish on the product singlet/central-triplet subspace because $L_{i,z} = I_{i,z} + I_{i',z}$ annihilates both states. A scalar identity term can be subtracted. Interpair nonsecular flip-flops are a separate approximation: their corrections require chemical-shift separations and individual couplings absent from the five-parameter row. The model assumes chemically equivalent protons within each pair, isotropic intrapair coupling, secular interpair coupling, and no RF during evolution.

Model	f12 / Hz	f23 / Hz	f13 / Hz	Max. shift from q=0 / Hz
Exact nearest-neighbor	1.967922603	2.185439927	4.153362531	0.000000000
Drop type-II (bare 3x3)	2.021632785	2.139776566	4.161409350	0.053710181
Second-order effective 3x3	1.967635409	2.185663097	4.153298506	0.000287194
Replace geminals by their mean	1.883537023	1.991550525	3.875087549	0.278274982
Fitted q, even parity	1.951897472	2.203534522	4.155431994	0.018094595
Fitted q, odd parity (matched)	1.983926090	2.167444914	4.151371004	0.017995013

Keep unequal geminals and type-II mixing. Averaging the geminals shifts a principal line by 0.278275 Hz. Dropping the fourth state shifts a line by 0.053710 Hz. Three observed principal peaks do not justify removing that state: weakly observed states still shift energies by virtual mixing. At this reference the fourth-basis-state probabilities in the three principal eigenstates are 0.257708963, 0.047940069, 0.203243205 percent.

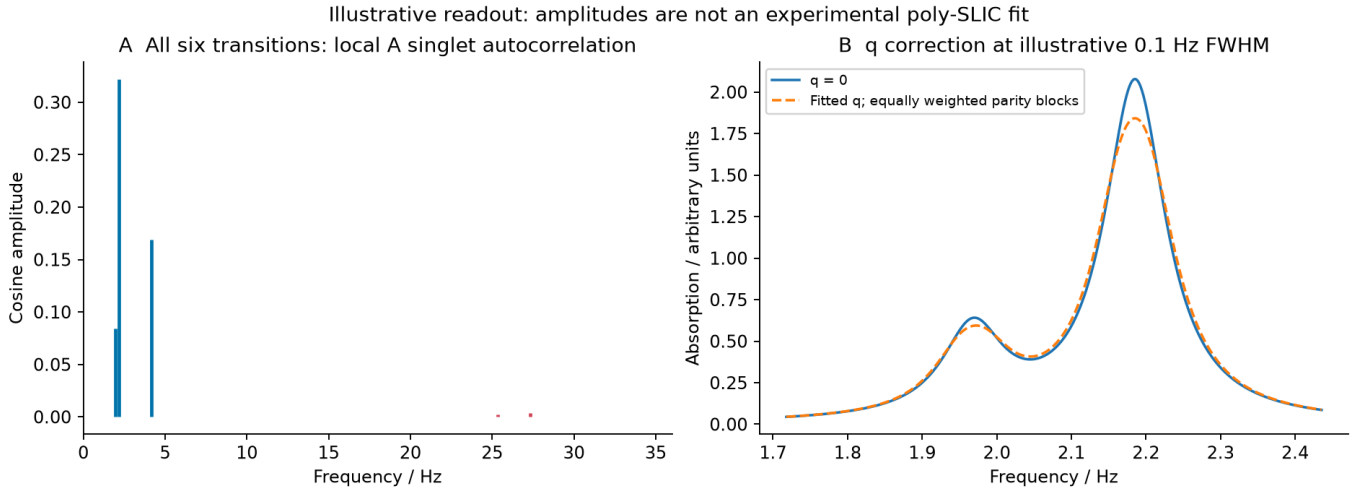
A controlled elimination writes $H = \begin{pmatrix} A & w \\ w^T & b \end{pmatrix}$. The exact energy-dependent reduced equation is $(A + ww^T/(E-b))\psi = E\psi$. In an eigenbasis of A, a Hermitian second-order approximation is

$$(H_{\mathrm{eff}})_{ij} = a_i\delta_{ij} + \frac{w_i w_j}{2}[(a_i - b)^{-1} + (a_j - b)^{-1}].$$

Its maximum principal-line error here is 0.000287194 Hz. It is much better than simply deleting the fourth state; keep its induced off-diagonal elements. The reference cubic below is exact within the q=0 block.

The q correction matters for precision. The supplied q changes a single-parity line by up to 0.018095 Hz and separates matched even/odd lines by up to **0.036089608 Hz**. They may overlap at a 0.1 Hz linewidth, but q should be included for millihertz line-center comparisons. The two parity blocks cannot automatically be represented by a single triplet when that splitting is resolved. Their relative preparation/readout weights are experiment-dependent. The first-order shifts have opposite signs in the matched parity blocks: an equally weighted pair of lines has a centroid shift only at second order. Thus the quoted single-parity shift is not automatically a shift of the maximum of an unresolved experimental peak; q can also alter its shape.

An analytical first-order correction is $E_i(q) = E_i(0) + q\langle i \mid Q \mid i\rangle + O(q^2)$, with $Q = \partial H/\partial q$ containing 1/2 at the four q positions. The principal-frequency derivatives at q=0 are **-0.586480376, 0.660847637, 0.074367261** (Hz per Hz). At the supplied q the first-order even-block frequencies are **1.951908170, 2.203485033, 4.155393203 Hz**, with maximum error **4.948893e-05 Hz** against full 4×4 diagonalization. This is a physics correction, not an extra learned output. A model that varies q independently needs a fifth normalized input.



3. Exact normalization, analytical eigenvalues, and branch labels

Use the same normalization as for DSS:

$$s = -J_M > 0, \quad E_{\mathrm{ref}} = (-3J_A + J_M - 3J_X)/4,$$

$$x_A = (J_A - J_M)/s, \quad x_X = (J_X - J_M)/s, \quad u = d/s, \quad v = e/s.$$

For homotaurine, s = 13.969277 Hz, E_{ref} = 16.64750575 Hz, and (x_A, x_X, u, v) = **(0.087717782388, -0.010014834698, 0.221443529254, 0.167824719919)**. Five dimensional parameters thus become four independent inputs. For q=0,

$$K = \frac{H - E_{\mathrm{ref}} I}{s} = \begin{pmatrix} x_A & u/2 & 0 & v/2 \\ u/2 & 0 & v/2 & 0 \\ 0 & v/2 & x_X & u/2 \\ v/2 & 0 & u/2 & -2 + x_A + x_X \end{pmatrix}.$$

Let $a = J_A / s = x_A - 1$, $c = J_X / s = x_X - 1$, and

$$B = \begin{pmatrix} a & u & 0 \\ 0 & -1 & v \\ 0 & 0 & c \end{pmatrix}.$$

The three ordered squared singular values of B are the roots $z_1 \le z_2 \le z_3$ of

$$z^3 - C_1 z^2 + C_2 z - C_3 = 0,$$

$$C_1 = a^2 + 1 + c^2 + u^2 + v^2,$$

$$C_2 = a^2 + a^2 c^2 + c^2 + a^2 v^2 + c^2 u^2 + u^2 v^2, \qquad C_3 = a^2 c^2.$$

For an explicit trigonometric expression, set $m = C_1 / 3$, $r = \sqrt{C_1^2 - 3C_2} / 3$, and

$$\theta = \frac{1}{3}\arccos\left(\frac{2C_1^3 - 9C_1 C_2 + 27C_3}{2(C_1^2 - 3C_2)^{3/2}}\right).$$

$$(z_1, z_2, z_3) = m + 2r(\cos(\theta + 2\pi/3), \cos(\theta - 2\pi/3), \cos\theta).$$

At r=0 all roots equal m. Numerical evaluation uses centered matrix invariants, with a symmetric eigensolver fallback very near repeated roots to avoid cancellation. This does not change the analytical identity. Define $\omega_i = s\sqrt{z_i}$ and $c_0 = -S_J / 4$. For negative geminals,

$$E_0 = c_0 - (\omega_1 + \omega_2 + \omega_3) / 2,$$

$$E_1 = c_0 + (\omega_1 + \omega_2 - \omega_3) / 2,$$

$$E_2 = c_0 + (\omega_1 - \omega_2 + \omega_3) / 2,$$

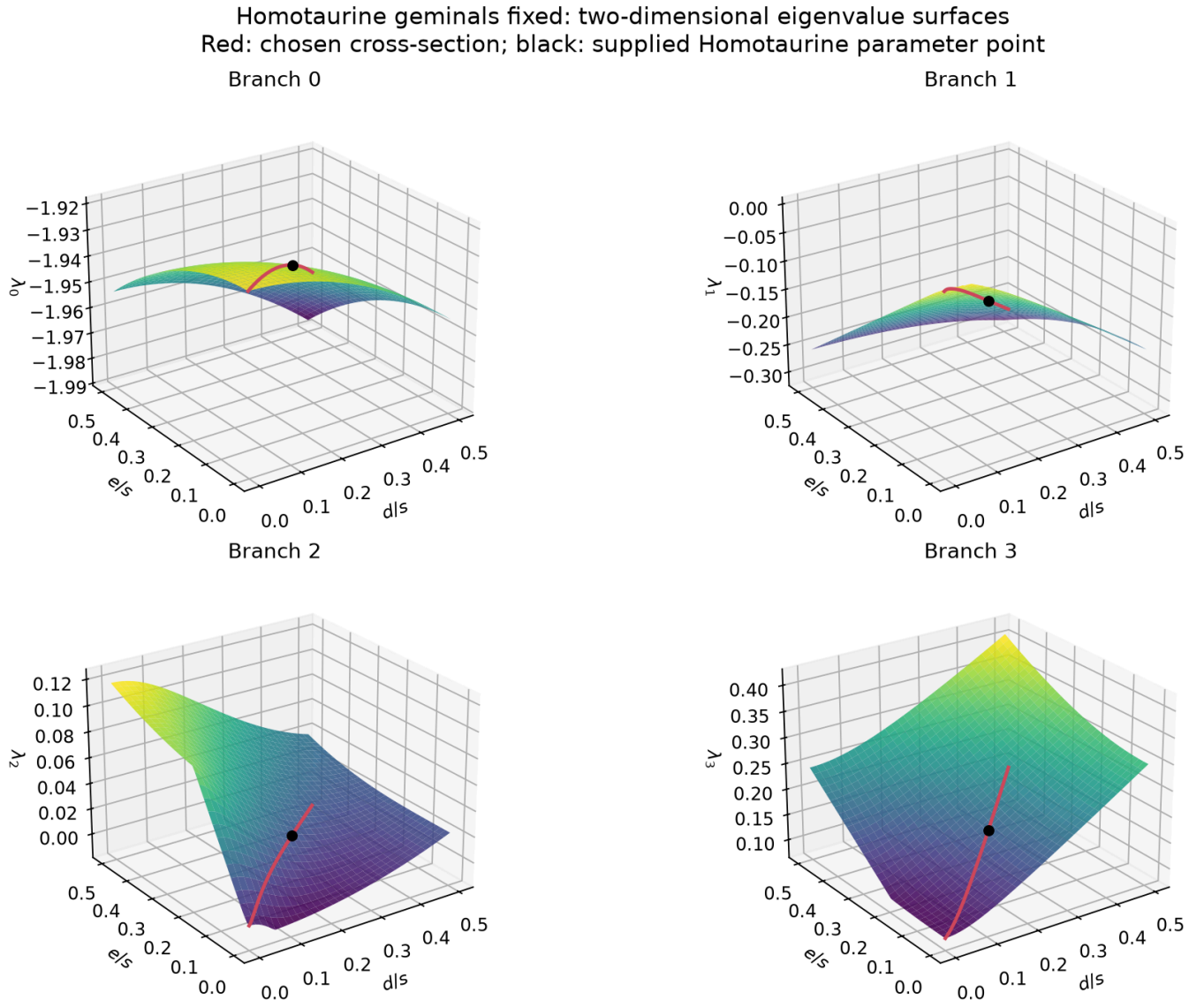
$$E_3 = c_0 + (-\omega_1 + \omega_2 + \omega_3) / 2.$$

These are ordered physical energies $E_0 \le E_1 \le E_2 \le E_3$. The supplied point has eigenvalues **-10.344029438, 15.015040435, 16.982963038, 19.168402965 Hz**, and normal-mode energies **11.586814973, 13.772254900, 15.740177504 Hz**. The principal transitions are

$$f_{12} = E_2 - E_1 = \omega_3 - \omega_2, \quad f_{23} = E_3 - E_2 = \omega_2 - \omega_1,$$

$$f_{13} = E_3 - E_1 = f_{12} + f_{23}.$$

Keep these index labels even if the two adjacent frequencies cross. Sorting their magnitudes would swap their identities. Ordered energy surfaces are continuous but may have cusps at degeneracy; a fixed ordered label is not a globally smooth eigenvector identity. Since s is positive, normalized energies have the same ascending order. All four targets are retained regardless of their sign.



4. Analytical and learned cross-sections of the same model

Fix the supplied geminals and put $d(t) = 3.093406t$ Hz and $e(t) = 2.344390t$ Hz. The physical reference is t=1; this is a mathematical parameter section, not a temperature calibration. Along it,

$C_1(t) = 2.852388810881 + 0.077202373264t^2$,

$C_2(t) = 2.701400997954 + 0.073465038627t^2 + 0.001381140469t^4$,

$C_3(t) = 0.849012187073$.

Insert these expressions into the ordered root formulas to obtain the exact analytical spectral section. Differentiating that solution at $\delta = t - 1 = 0$ gives the following Taylor polynomials, with omitted terms $O(\delta^3)$:

$f_{12} = -0.065824439\delta^2 + 1.787864456\delta + 1.967922603$ Hz.

$f_{23} = 0.277518001\delta^2 + 1.814718688\delta + 2.185439927$ Hz.

$f_{13} = 0.211693562\delta^2 + 3.602583144\delta + 4.153362531$ Hz.

Their sampled maximum error on $0.9 \le t \le 1.1$ is **0.0002443686 Hz**. The derivatives are calculated analytically with Hermitian eigenvalue derivative identities; these are not fitted polynomials.

Substitution of this section into the **actual trained four-input ASN** gives:

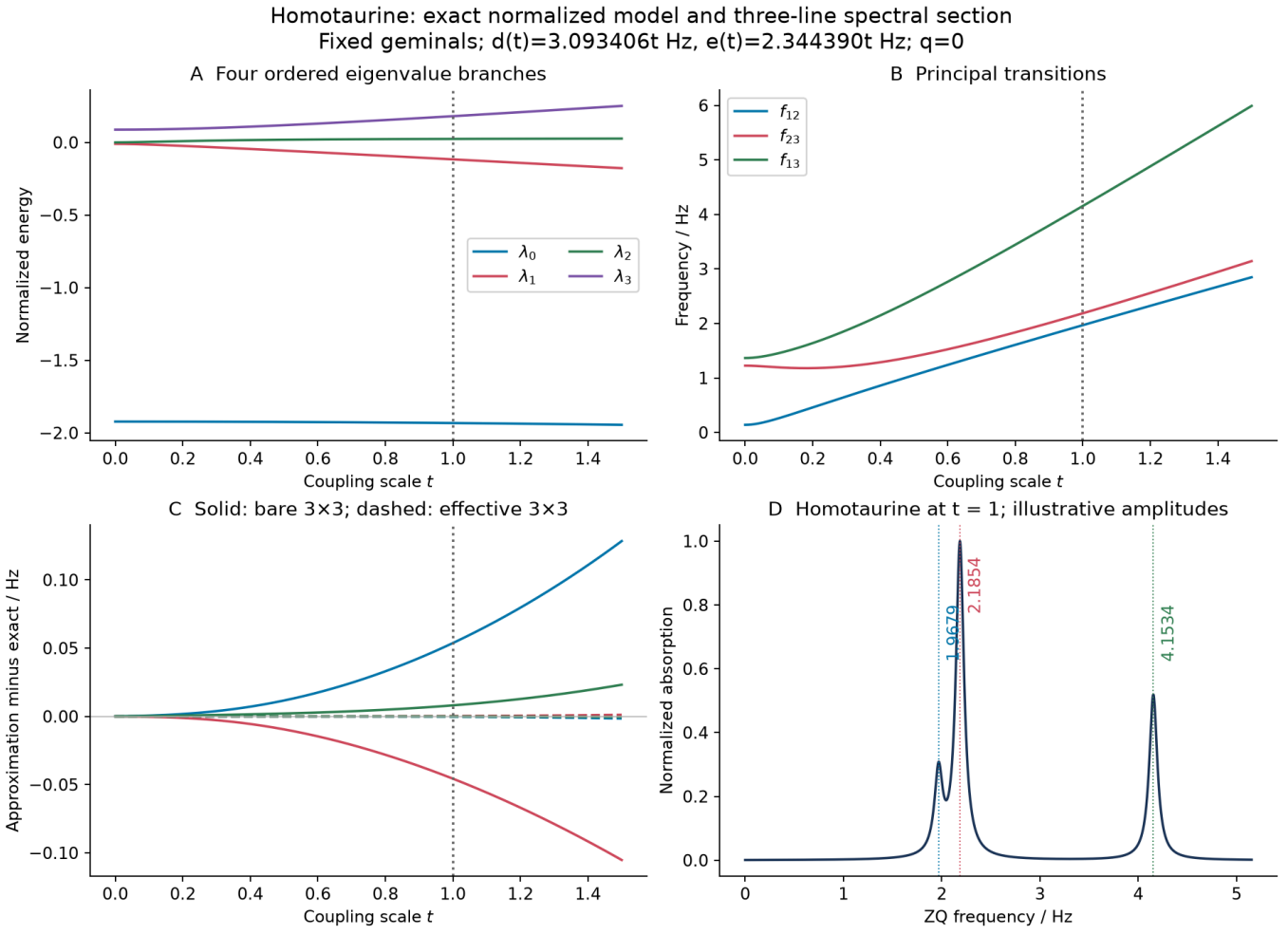
$$\hat{f}_{12} = -0.062329221t^2 + 1.904871944t + 0.12563094 \quad \text{Hz}.$$

$$\hat{f}_{23} = 0.281058571t^2 + 1.251177692t + 0.653152603 \quad \text{Hz}.$$

$$\hat{f}_{13} = 0.21872935t^2 + 3.156049636t + 0.778783543 \quad \text{Hz}.$$

These expressions are composed from learned layer weights and activation mixtures. No one-dimensional fitting was performed. Full coefficients are saved; nine decimals are shown here. They approximate a finite training domain and should not be identified with the exact Taylor coefficients. On the local 161-point section $0.8 \le t \le 1.2$, their frequency RMSE is **0.0007485736 Hz** and maximum error **0.002169341 Hz**. The derivative comparison about t=1 is:

Line	Exact slope	ASN slope	Exact quadratic coefficient	ASN quadratic coefficient
f12	1.787864456	1.780213501	-0.065824439	-0.062329221
f23	1.814718688	1.813294834	0.277518001	0.281058571
f13	3.602583144	3.593508336	0.211693562	0.218729350



5. Actual training and extracted four-input formulas

The repository class `nmrkan.models.KharKAN` implements the manuscript's ASN with activation choices $\{0,x,x^2\}$; this is not a spline-edge KAN. We trained (4,16,4) networks with biases using Adam, learning rate 0.001, batch size 4096, and 1000 epochs. Epochs 1–900 use softmax weights and a 0.1×L0.5 regularizer; epochs 901–1000 use sparsemax without the regularizer. The loss is MAE on training-standardized targets. Affine input and output conditioning is fitted only to training data and undone algebraically in the extracted formulas. This conditioning is an explicit extension of the original manuscript's protocol, matching the DSS experiment.

The **local domain** independently varies each homotaurine geminal by ±0.3 Hz and each coupling difference by ±20%. The **broad domain** uses the same original box as DSS: geminals in [−15,−12.5] Hz and differences in [0.1,7] Hz. The local J_A upper bound is −12.443923 Hz, so the local domain is not wholly contained in the broad box. These are numerical approximation domains, not parameter confidence intervals.

Each domain uses 100,000 vectors generated with NumPy seed 42, split into 80,000 training, 10,000 validation, and 10,000 test vectors. Five initialization/minibatch seeds (42–46) are trained. Checkpoint and seed selection minimize unstandardized dimensionless eigenvalue RMSE on validation samples, restricted to the sparsemax phase. Test evaluation follows selection. The exact reference vector and section were not deliberately inserted in training. The broad run uses the same physical distribution and seeds as DSS: it is the same general Hamiltonian benchmark evaluated on a different molecular section, not an independent second-molecule training population.

The selected local model is seed **45**, epoch **971**. Extracted degrees are **[2, 2, 2, 2]**. Numerical targets come only from `numpy.linalg.eigvalsh` on the physical 4×4 block, followed by normalization. The model never receives cubic invariants, singular modes, or analytical derivatives. We impose no post-prediction sorting, positivity repair, symmetry averaging, or trace projection.

The following coefficient table defines $\hat{\lambda}_i = \sum_m c_{im} m(x_A, x_X, u, v)$ after the manuscript's presentation step: discard coefficients with magnitude below 0.001 and round remaining coefficients to three decimals. The main performance results use the separate full-precision formulas.

Monomial	λ0	λ1	λ2	λ3
1	-2.000	0.000	0.000	0.000
v	0.002	-0.402	0.241	0.168
u	0.001	-0.289	-0.160	0.447
xX	1.002	0.306	0.618	0.076
xA	1.004	0.152	0.286	0.560
v^2	-0.128	-0.376	-0.072	0.547
uv	0.000	1.043	-0.899	-0.131
u^2	-0.128	-0.586	0.619	0.092
vxX	-0.025	1.259	-2.138	0.879
uxX	0.000	-1.509	1.755	-0.241
xX^2	0.000	-1.335	0.916	0.415
vxA	-0.007	-0.805	2.024	-1.235
uxA	-0.031	1.106	-0.952	-0.129
$xAxX$	0.002	0.914	-0.044	-0.861
xA^2	0.001	-0.576	-0.723	1.311

Full expressions are in `homotaurine_asn_comparison/local/learned_formulas.json` and `.tex`. Their maximum discrepancy from the double-precision trained network on 1000 validation points is **3.55e-15**. Checkpoint reload and formula agreement are also checked on held-out samples. The three-decimal formulas yield **1.964987609, 2.171449967, 4.136437576 Hz** at the reference; their distinct error row below quantifies the cost of rounding.

6. Held-out benchmarks and branch-order checks

Each method within a domain uses the same test samples. Complete quadratic (15 monomials) and quartic (70 monomials) ordinary least-squares fits use the same 80,000 training samples. Their squared-error objective differs from the ASN's MAE/regularization objective. The analytical cubic, effective 3×3, and bare 3×3 are physics references with no fitted coefficients.

Homotaurine local domain

Method	Eigenvalue RMSE / Hz	Frequency RMSE / Hz	Max. line error / Hz	Ordering violations
Exact analytical cubic	4.52299e-15	5.37437e-15	2.4869e-14	0
ASN	0.00107861	0.00213516	0.0210831	0
ASN paper-style 3-decimal rounding	0.00692589	0.0130434	0.0302093	0
OLS degree 2	0.00102477	0.0020495	0.0187465	0
OLS degree 4	2.97338e-05	5.94677e-05	0.00147479	0
Effective 3x3	—	0.000229722	0.00070852	—
Bare 3x3	—	0.0415378	0.0940144	—

Broad supplied domain

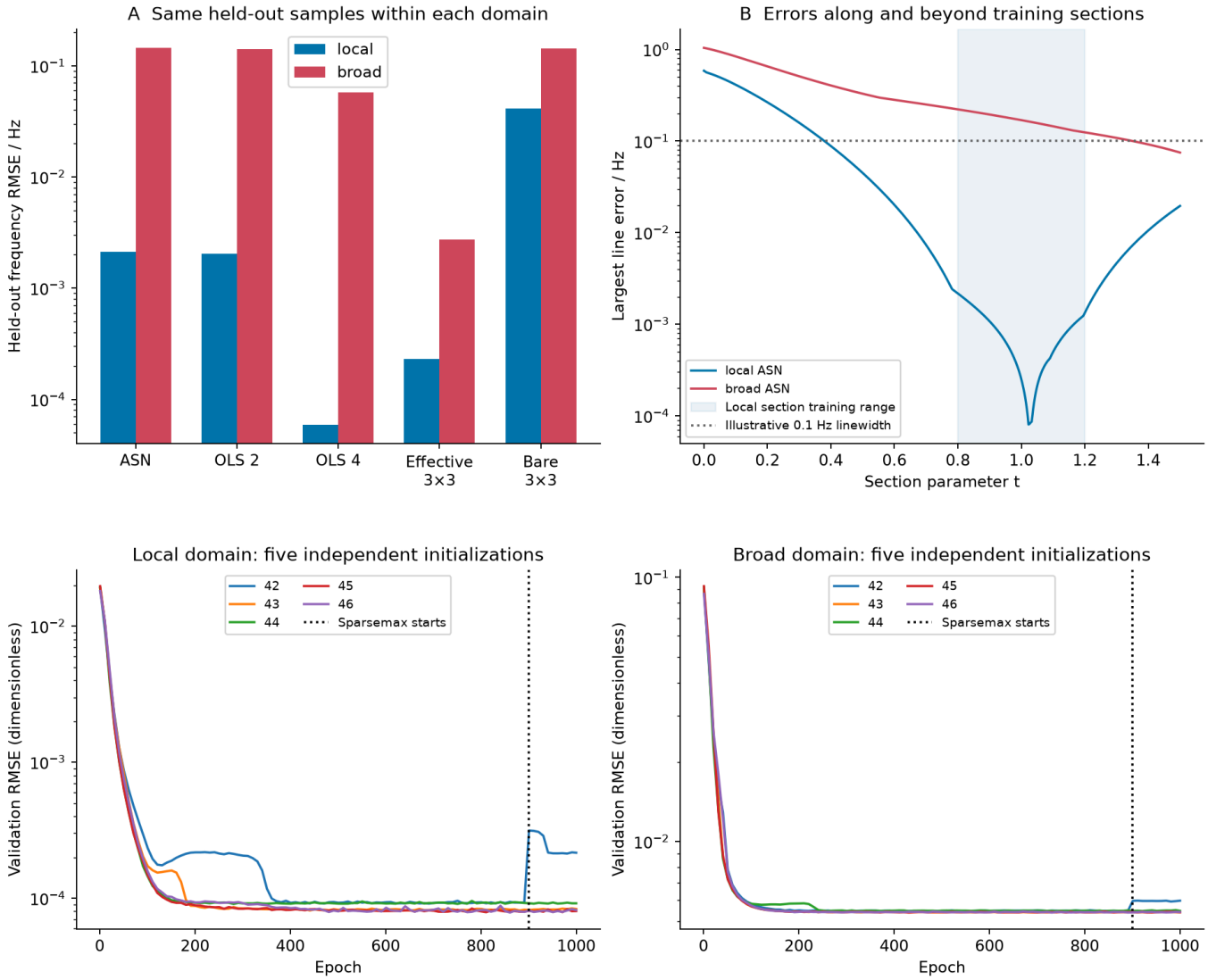
Method	Eigenvalue RMSE / Hz	Frequency RMSE / Hz	Max. line error / Hz	Ordering violations
Exact analytical cubic	5.3134e-15	6.71216e-15	4.30767e-14	0
ASN	0.0725538	0.144956	1.92457	4
ASN paper-style 3-decimal rounding	0.0725919	0.145038	1.91998	4
OLS degree 2	0.0703782	0.140741	1.72007	1
OLS degree 4	0.0288095	0.0576189	0.711797	0
Effective 3x3	—	0.00273032	0.020747	—
Bare 3x3	—	0.142218	0.55397	—

The broad model predicts **2.137294453, 2.026477047, 4.163771501 Hz** at homotaurine. Its global test error must not be confused with local approximation accuracy. Seed variability is shown below; all seeds share the same dataset, so this measures optimization variation rather than experimental uncertainty.

Domain	Seed	Selected epoch	Validation energy RMSE (dimensionless)	Test frequency RMSE / Hz
local	42	981	0.0002127109	0.005706662
local	43	901	8.176663e-05	0.00212556
local	44	911	9.107266e-05	0.002146188
local	45 (selected)	971	7.887293e-05	0.002135162
local	46	951	7.898869e-05	0.002152738
broad	42	981	0.005932914	0.1520699
broad	43	941	0.005397352	0.1449643
broad	44	921	0.005461121	0.1465545
broad	45 (selected)	931	0.005391291	0.1449562
broad	46	971	0.005394605	0.1447796

The selected local and broad models have **0** and **4** ordering violations, respectively, on their 10,000 test points. A separate 2000-point challenge with broad-box geminals sets d=0 or e=0; these are outside both sampled boxes and probe the decoupled boundary. Its local/broad maximum frequency errors are **4.220976234, 2.005351419 Hz**, with **452, 52** violations. We do not sort predictions afterward to hide these failures. Zero sampled violations would still not prove global ordering.

The learned identity $\hat{f}_{13} = \hat{f}_{12} + \hat{f}_{23}$ holds by construction. Trace errors, per-seed statistics, boundary metrics and all configurations are retained in JSON. Broad-domain ordering cusps are an intrinsic limitation of a finite smooth polynomial representation.



7. From three energy gaps to an observable spectrum

For preparation ρ and readout O, the signal is $S(\tau) = \mathrm{Tr}[Oe^{-i2\pi H\tau}\rho e^{i2\pi H\tau}]$. Each energy pair contributes a cosine amplitude $2\mathrm{Re}(\rho_{ij}O_{ji})$ and sine amplitude $-2\mathrm{Im}(\rho_{ij}O_{ji})$. A frequency need not be visible merely because an energy difference exists.

For a reproducible illustration we use the local-A singlet deviation operator $\rho = O = P_{S,A} - I/2 = \mathrm{diag}(-1/2, 1/2, 1/2, -1/2)$ in the even block. It describes a deviation from background, not a positive density operator by itself. At q=0 the three principal cosine amplitudes in f12,f23,f13 order are 0.084349, 0.322096, and 0.169245. Three additional weak transitions occur near 25.3591, 27.3270, and 29.5124 Hz. They are not forbidden; their illustrative amplitudes are 0.002392, 0.003910, and 0.000375.

The main comparison plots use the same three principal amplitudes and 0.1 Hz Lorentzian FWHM for both methods, replacing only the frequencies with ASN predictions. The separate full-spectrum CSV retains all six lines and both parity blocks for q=0 and fitted q. Equal parity weighting in the q illustration is an assumption; actual poly-SLIC preparation and reconversion must determine it. The ASN learns eigenvalues, not eigenvectors or pulse efficiencies. It does not predict experimental intensities or relaxation.

8. Paper narrative and reproducibility

The molecular demonstration now follows the full sequence requested by the editor: supplied real-molecule couplings, justified reduced Hamiltonian, normalization, numerical supervised targets, actual ASN training, extraction of explicit formulas, and comparison of observable line centers with an independent analytical solution. Retain the polynomial and effective-Hamiltonian baselines when interpreting accuracy. The results establish a useful learned approximation over a stated domain; they do not establish superiority over an appropriate polynomial basis or an exact small-matrix solver.

Homotaurine also exposes a physical-model limitation: a small surrogate error does not remove the omitted q correction or resolve the supplied fit's temperature ambiguity. The analytical and learned q=0 results are valid computational benchmarks. Experimental validation requires the final physical parameter row and measured spectrum, together with the preparation/readout model. The editor's separate request for a contemporary symbolic-regression baseline such as PySR/EQL remains separate; none is claimed here. Any manuscript statement that AI was not used to produce research calculations should be reconciled with the use of these artifacts.

- `homotaurine_asn_comparison/homotaurine_manuscript_addition.tex`: manuscript insertion with measured benchmark results.
- `homotaurine_asn_comparison/analytical_results.json`: exact energies, approximations, q correction, and Taylor coefficients.
- `homotaurine_asn_comparison/{local,broad}/`: datasets, split indices, configurations, hashes, five checkpoints per domain, training histories, full/rounded formulas, held-out predictions and section CSVs.
- Figures are saved as PDF, SVG and PNG; this HTML and the notebook embed images for portability.

Reproduce from the repository root:

```
python ANSWER/train_dss_asn.py --molecule homotaurine --domain local
python ANSWER/train_dss_asn.py --molecule homotaurine --domain broad
python ANSWER/build_homotaurine_report.py
python -m unittest discover -s ANSWER -p 'test*.py' -v
```

CPU training uses one computation thread per process; dependency versions and source/data hashes are recorded in each config. No repository model source was changed. Full-precision machine-readable artifacts should be used for numerical evaluation; printed coefficients are rounded.